

Project 1 — Testing a Claim

Is this observation surprising?

Project 1: Testing a hypothesis from a binomial random process

Total points: 35

Due: See [course path](#) and Canvas for Module 1 project due date.

In this project, you will write a **one-page summary** of how you used the **binomial distribution** to draw a conclusion about an infinite process from data you collected. For the inference to be valid, your data must satisfy the four conditions of a **binomial random process (BRP)**.

Sample write-up: [Download sample one-page write-up — ESP binomial test](#). This is my completed report for an ESP study. **Do not copy my process** (ESP is a class example you may not use for your own project). Write up **your own study** following the grading rubric below.

Milestone (precheck): [Project 1 milestone](#) · [milestone notebook](#) — due **before** final data collection; see [Module 1 schedule](#).

Project workflow

Phase	What	Where	Canvas
A — Milestone	Plan question, p , H_0/H_a , null graph (no x yet)	Milestone notebook · milestone hub	Milestone assignment
B — Full project	Collect $n = 25$, analyze, one-page report	Steps B1–B5 below + rubric	Project 1 assignment

Due dates: [course path](#) (milestone before project).

Steps

Phase A — Milestone precheck (before data collection)

Complete in the [milestone notebook](#) and submit in Canvas **before** you finish collecting data. Details: [milestone hub](#).

Step	Task
A1	Choose a binomial process with $n = 25$ (not a class example). Check all four BRP conditions: (a) binary outcome per trial, (b) independent trials, (c) fixed p per trial, (d) fixed n .
A2	Write your research question (one or two sentences).

Step	Task
A3	Define p in context — what counts as one “success” on a single trial.
A4	State H_0 and H_a with a numeric p in H_0 ($0 < p < 1$).
A5	Plot the null binomial distribution in R (<code>dbinom + geom_col</code>) using $n = 25$ and p from H_0 . Do not mark an observed x yet.

Code help: [Module 1 demo Section C](#) · [R reference](#).

Phase B — Full project (after milestone)

1. **OBSERVE** $n = 25$ trials. Record successes (x). Compute $\hat{p} = x/n$.
2. In **R**, plot the null distribution (same as milestone) and add a **descriptive** bar or pie chart for your observed counts.
3. Mark observed x on the null plot with `geom_vline(xintercept = ...)`.
4. Compute the **p-value** with `pbinom()` (match the direction of H_a). Interpret in context.
5. **Write up** a one-page summary using the **rubric below** (Introduction, Data Collection, Data Analysis, Conclusion).
Code help: [Module 1 Read 2](#) · [demo lab](#) · [R reference](#).

Project 1: One page summary of binomial test — grading rubric

Total points: 35

Format (2 pts)

Your summary should be organized into 4 sections (1 section with 2 subsections) with the following headings:

1. Introduction
2. Data Collection
3. Data Analysis
 - 3a. Descriptive Statistics
 - 3b. Inferential Statistics
4. Conclusion

Style (3 pts)

- Your summary should fit within **one side** of an 8.5 by 11 inch page. (1 point)
- Your summary should be concise and written in complete sentences with proper grammar. (2 points)

Introduction section (5 pts)

In this section, you should:

1. Briefly describe what question you are trying to answer. (1 point)
2. Briefly describe the motivation you have for trying to answer the question. (1 point)
3. Describe the parameter of interest p in one sentence. (1 point)

4. State the null and alternative hypotheses for your research question using appropriate notation and a value for p . (2 points)

Data collection section (7 pts)

- Describe **how, when, and where** you collected data in enough detail that someone else could repeat your study. (3 points)
- Explicitly explain why you believe **each of the four BRP conditions** was satisfied by your method. (4 points)
Explain in the **context of your study**, not as a generic list of the four conditions.

Descriptive statistics subsection (6 pts)

- Include the total number of trials n , the number of successes x , and the **percentage** of successes you observed. (3 points)
- Include a **well-labelled** pie chart or bar chart of your observed data, made in **R** (e.g. with `ggplot2`). (3 points)

Inferential statistics subsection (8 pts)

In this subsection, please include:

- A graph of the **binomial distribution** for all possible values of X using the p from H_0 . Create this as a **column chart in R** using `dbinom()` for the bar heights and `geom_col()`. (4 points)
- The **p-value**: the probability of seeing a new X as extreme as your x if H_0 is true. Compute it in R with `pbinom()` (or an equivalent correct method). (2 points)
- A brief explanation of **how** you calculated the p-value. (1 point)
- A clear indication of **where your data** x appears on the binomial graph (e.g. a vertical line or label). (1 point)

See the [ESP sample PDF](#) for an example of wording and figures.

Conclusion section (4 pts)

- Write a **one-sentence** summary of what your p-value means in the context of your research question and data. (2 points) (See the [sample project](#) for an example.)
- Based on the size of this p-value, state what you conclude about your research question. (2 points)